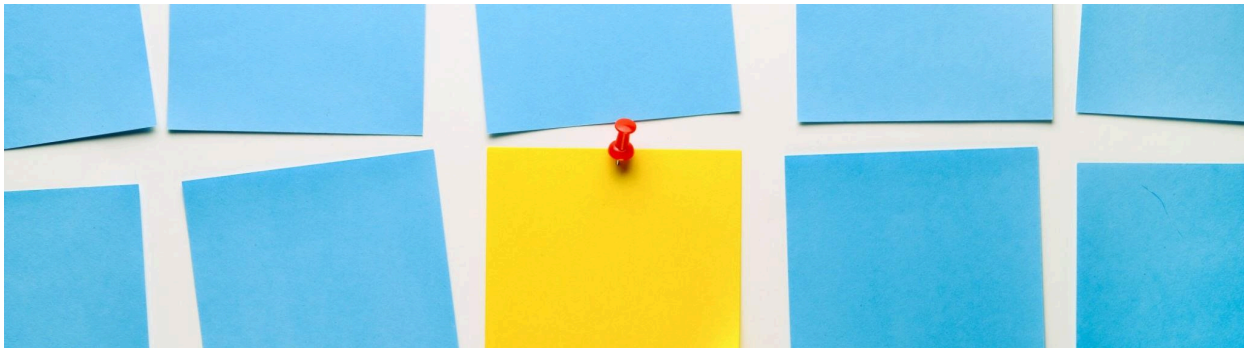

Data Preprocessing

Status Completed

Timing Mar 27, 2026 to Mar 27, 2026

Owners Krisha Anghan

Overview

The dataset contains patient-related information such as age, gender, region, BMI, blood pressure, cholesterol, and glucose levels. It also includes a target variable called disease_risk, which indicates whether a patient has a low or high health risk.

Objectives

The main objective of this project is to improve data quality by identifying and handling missing values, detecting and treating

outliers, comparing different preprocessing methods, and preparing the dataset for machine learning.

Simple Imputer

Simple Imputer is a basic technique used to fill missing values using statistical measures. For numerical data, it replaces missing values with the mean or median, while for categorical data, it uses the most frequent value. It is easy to implement but may not always preserve data patterns.

Most Frequent Imputation

This method replaces missing values with the most common value (mode) in a column. It is mainly used for categorical features and helps maintain consistency in the dataset.

Missing Indicator And Random Sample Imputation

In this technique, a new column is created to indicate whether a value was missing or not. Then, missing values are filled by randomly selecting values from existing data in that column. This helps in preserving the original data distribution.

KNN Imputer

K-Nearest Neighbors (KNN) Imputer fills missing values based on the similarity between data points. It finds the nearest neighbors and uses their values to estimate the missing ones. This method provides more realistic results compared to simple methods.

MICE Algorithm

MICE (**Multiple Imputation by Chained Equations**) is an advanced imputation technique that fills missing values by modeling each feature as a function of other features. It performs multiple iterations to generate accurate estimates and captures relationships between variables effectively.

Z-score Method

The Z-score method is used to identify outliers by measuring how far a data point is from the mean in terms of standard deviations. Values that fall beyond a certain threshold (usually ±3) are considered outliers. This method works well when the data follows a normal distribution.

IQR (Interquartile Range) Method

The IQR method detects outliers based on the spread of the middle 50% of the data. It calculates the difference between the first quartile (Q1) and the third quartile (Q3). Any value below Q1 − 1.5×IQR or above Q3 + 1.5×IQR is treated as an outlier. This method is robust and works well for skewed data.

Percentile Method

The percentile method handles outliers by setting limits based on percentiles. For example, values below the 1st percentile and above the 99th percentile are considered extreme and can be removed or capped. This method ensures that only a small portion of extreme data is affected.

Winsorization Technique

Winsorization is used to limit extreme values by capping them at a specific percentile range instead of removing them. For example, values below the lower limit are replaced with the minimum threshold, and values above the upper limit are replaced with the maximum threshold. This helps in reducing the impact of outliers while preserving all data points.